

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Civil) (Credit System) Degree Examinations**

January 2023

20CV505 – ENVIRONMENTAL ENGINEERING - I

Duration: 3 Hours.

Max. Marks: 100

- Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.
 2) Missing data may be suitably assumed.

Unit – I**Marks BT* CO* PO***

1. a) The census data of population of a town are as follows. Estimate the population by the year 2031 by Arithmetic Increase Method, Geometric Increase Method and Incremental Increase Method.

Year	1991	2001	2011	2021
Population (in thousands)	80	120	145	160

Estimate the important kinds of draft which may be required to be recorded for an average consumption of 250lpcd.

10 L*3 1 2

- b) List the different types of intake structures based on the source of water. Illustrate and explain any one type.

10 L2 2 1

2. a) Calculate the required capacities of the major components of a proposed water works system for a city. The city uses a river as the source of water supply for a total population of 200000. Assume an average daily demand of 50MLD.

10 L3 1 2

- b) Write a short note on the health significance of (i) Fluorides, (ii) Nitrates, (iii) Mercury, and (iv) Cadmium in drinking water. Mention their acceptable and permissible limits in drinking water as per IS 10500: 2012.

10 L1 2 8

3. a) Illustrate a typical flow chart of water supply scheme. Outline the points to be considered for a public water supply scheme.

10 L3 1 6

- b) Design a set of circular settling tanks to treat 5 MLD of water. Take the detention period as 2 hours. Calculate the overflow rate and weir loading.

10 L3 2 3

Unit – II

4. a) Explain with a neat sketch the mechanism of the electrostatic stabilization of colloids using the Electrical Double Layer Theory.

10 L2 3 1

- b) Illustrate and explain the salient points in a breakpoint chlorination curve

10 L3 4 2

5. a) The water works of Mangalore city treats 50MLD in a coagulation-sedimentation tank. The quantity of alum consumed is 15 mg/L. If the alkalinity of raw water is equivalent to 1mg/L as CaCO₃, calculate the quantity of alum and quick lime (containing 85% CaO) required per year by the water works. Take the atomic weights of Ca, S, Al, O, C, and H as 40, 32, 27, 16, 12, and 1 respectively.

10 L3 3 2

- b) Explain with a neat sketch the method of removal of hardness by lime-soda process.

10 L2 4 1

6. a) Explain in brief the different processes through which water gets purified in the filtration process.

10 L2 3 1

P.T.O.

- b) Water must be supplied to a community with an average daily demand of 3MLD. The disinfectant used for chlorination is bleaching powder containing 30% of available chlorine. Calculate the quantity of bleaching powder required annually at the water works if 0.3 mg/L Chlorine is required for disinfection.

10 L3 4 2

Unit – III

7. a) List and explain the different methods for removal of iron and manganese from water.
 b) Enumerate the disadvantages of intermittent water supply system.
8. a) Explain in brief the function of activated carbon treatment of water.
 b) Explain with neat sketches the different methods of water distribution systems.

10 L2 5 1
 10 L2 5 6
 10 L2 5 1
 10 L2 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
